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Parallel Computing Assignment 1

N	Mode	Threads	Time (s)	Speedup	Efficiency	Schedule
512	0	1	0.240595	1	100	Serial
512	1	1	0.238863	1	100	parallel for, schedule(static)
512	1	2	0.126378	1.89	95	parallel for, schedule(static)
512	1	4	0.067587	3.53	88	parallel for, schedule(static)
512	1	8	0.039172	6.1	76	parallel for, schedule(static)
512	1	12	0.029433	8.11	68	parallel for, schedule(static)
512	1	16	0.02914	8.2	51	parallel for, schedule(static)
512	1	20	0.02624544	9.1	46	parallel for, schedule(static)
512	2	1	0.237231	1	100	parallel for, collapse(2), schedule(static)
512	2	2	0.12467	1.9	95	parallel for, collapse(2), schedule(static)
512	2	4	0.068075	3.48	87	parallel for, collapse(2), schedule(static)
512	2	8	0.039172	6.06	76	parallel for, collapse(2), schedule(static)
512	2	12	0.033287	7.13	59	parallel for, collapse(2), schedule(static)
512	2	16	0.025621	9.26	58	parallel for, collapse(2), schedule(static)
512	2	20	0.034409778	6.89	34	parallel for, collapse(2), schedule(static)
512	3	1	0.238274	1	100	parallel for, schedule(static) + atomic (checksum)
512	3	2	0.123893	1.92	96	parallel for, schedule(static) + atomic (checksum)
512	3	4	0.066838	3.57	89	parallel for, schedule(static) + atomic (checksum)
512	3	8	0.039108	6.09	76	parallel for, schedule(static) + atomic (checksum)
512	3	12	0.03161	7.54	63	parallel for, schedule(static) + atomic (checksum)
512	3	16	0.02724	8.75	55	parallel for, schedule(static) + atomic (checksum)
512	3	20	0.027299889	8.73	43	parallel for, schedule(static) + atomic (checksum)
512	4	1	0.241161	1	100	parallel for, schedule(static) + critical (checksum)

512	4	2	0.123108	1.96	98	parallel for, schedule(static) + critical (checksum)
512	4	4	0.068424	3.52	88	parallel for, schedule(static) + critical (checksum)
512	4	8	0.040236	5.99	75	parallel for, schedule(static) + critical (checksum)
512	4	12	0.035111	6.87	57	parallel for, schedule(static) + critical (checksum)
512	4	16	0.027158	8.88	55	parallel for, schedule(static) + critical (checksum)
512	4	20	0.024108556	10.01	50	parallel for, schedule(static) + critical (checksum)
512	5	1	0.237513	1	100	task-based parallelism (blocked row decomposition, block_size=16)
512	5	2	0.123164	1.93	97	task-based parallelism (blocked row decomposition, block_size=16)
512	5	4	0.068432	3.47	87	task-based parallelism (blocked row decomposition, block_size=16)
512	5	8	0.037974	6.26	78	task-based parallelism (blocked row decomposition, block_size=16)
512	5	12	0.028384	8.37	70	task-based parallelism (blocked row decomposition, block_size=16)
512	5	16	0.024648	9.64	60	task-based parallelism (blocked row decomposition, block_size=16)
512	5	20	0.025947444	9.15	46	task-based parallelism (blocked row decomposition, block_size=16)
512	6	1	0.238965	1	100	parallel for, schedule(static) + simd (inner k-loop vectorization)
512	6	2	0.124031	1.93	97	parallel for, schedule(static) + simd (inner k-loop vectorization)
512	6	4	0.067093	3.56	89	parallel for, schedule(static) + simd (inner k-loop vectorization)
512	6	8	0.037981	6.29	79	parallel for, schedule(static) + simd (inner k-loop vectorization)
512	6	12	0.031143	7.67	64	parallel for, schedule(static) + simd (inner k-loop vectorization)
512	6	16	0.025185	9.49	59	parallel for, schedule(static) + simd (inner k-loop vectorization)
512	6	20	0.027283222	8.76	44	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	0	1	1.927162	1	100	Serial
1024	1	1	1.895679	1	100	parallel for, schedule(static)
1024	1	2	1.010550444	1.88	94	parallel for, schedule(static)
1024	1	4	0.531021111	3.57	89	parallel for, schedule(static)

1024	1	8	0.282415889	6.71	84	parallel for, schedule(static)
1024	1	12	0.302584333	6.27	52	parallel for, schedule(static)
1024	1	16	0.243361667	7.79	49	parallel for, schedule(static)
1024	1	20	0.233994889	8.1	41	parallel for, schedule(static)
1024	2	1	1.899000778	1	100	parallel for, collapse(2), schedule(static)
1024	2	2	0.989667556	1.92	96	parallel for, collapse(2), schedule(static)
1024	2	4	0.524253222	3.62	90	parallel for, collapse(2), schedule(static)
1024	2	8	0.284813556	6.67	83	parallel for, collapse(2), schedule(static)
1024	2	12	0.304807556	6.23	52	parallel for, collapse(2), schedule(static)
1024	2	16	0.240347333	7.9	49	parallel for, collapse(2), schedule(static)
1024	2	20	0.248757333	7.63	38	parallel for, collapse(2), schedule(static)
1024	3	1	1.903410222	1	100	parallel for, schedule(static) + atomic (checksum)
1024	3	2	0.993037333	1.92	96	parallel for, schedule(static) + atomic (checksum)
1024	3	4	0.530691778	3.59	90	parallel for, schedule(static) + atomic (checksum)
1024	3	8	0.285098	6.68	84	parallel for, schedule(static) + atomic (checksum)
1024	3	12	0.304923111	6.24	52	parallel for, schedule(static) + atomic (checksum)
1024	3	16	0.239984889	7.93	50	parallel for, schedule(static) + atomic (checksum)
1024	3	20	0.238333111	7.99	40	parallel for, schedule(static) + atomic (checksum)
1024	4	1	1.899709778	1	100	parallel for, schedule(static) + critical (checksum)
1024	4	2	1.019883889	1.86	93	parallel for, schedule(static) + critical (checksum)
1024	4	4	0.526594	3.61	90	parallel for, schedule(static) + critical (checksum)
1024	4	8	0.284226	6.68	84	parallel for, schedule(static) + critical (checksum)
1024	4	12	0.305065778	6.23	52	parallel for, schedule(static) + critical (checksum)
1024	4	16	0.245313222	7.75	48	parallel for, schedule(static) + critical (checksum)
1024	4	20	0.235663889	8.06	40	parallel for, schedule(static) + critical (checksum)
1024	5	1	1.900686111	1	100	task-based parallelism (blocked row decomposition, block_size=16)
1024	5	2	1.149739667	1.65	83	task-based parallelism (blocked row decomposition, block_size=16)

1024	5	4	0.527697111	3.6	90	task-based parallelism (blocked row decomposition, block_size=16)
1024	5	8	0.285220222	6.67	83	task-based parallelism (blocked row decomposition, block_size=16)
1024	5	12	0.255731667	7.43	62	task-based parallelism (blocked row decomposition, block_size=16)
1024	5	16	0.195606444	9.72	61	task-based parallelism (blocked row decomposition, block_size=16)
1024	5	20	0.19965778	9.52	48	task-based parallelism (blocked row decomposition, block_size=16)
1024	6	1	1.886810444	1	100	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	6	2	0.991822889	1.9	95	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	6	4	0.521867556	3.61	90	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	6	8	0.286990444	6.58	82	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	6	12	0.301007	6.27	52	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	6	16	0.242838778	7.77	49	parallel for, schedule(static) + simd (inner k-loop vectorization)
1024	6	20	0.236830111	7.97	40	parallel for, schedule(static) + simd (inner k-loop vectorization)
2048	0	1	42.501402	1	100	Serial
2048	1	1	49.07217833	1	100	parallel for, schedule(static)
2048	1	2	28.48810511	1.72	86	parallel for, schedule(static)
2048	1	4	11.22965933	4.37	109	parallel for, schedule(static)
2048	1	8	5.275286444	9.3	116	parallel for, schedule(static)
2048	1	12	5.557290333	8.83	74	parallel for, schedule(static)
2048	1	16	4.288917556	11.44	71	parallel for, schedule(static)
2048	1	20	4.04535	12.13	61	parallel for, schedule(static)
2048	2	1	47.29707889	1	100	parallel for, collapse(2), schedule(static)
2048	2	2	27.86990789	1.7	85	parallel for, collapse(2), schedule(static)
2048	2	4	10.94963711	4.32	108	parallel for, collapse(2), schedule(static)
2048	2	8	5.484049778	8.63	108	parallel for, collapse(2), schedule(static)
2048	2	12	5.553293556	8.51	71	parallel for, collapse(2), schedule(static)

2048	2	16	4.306929556	10.98	69	parallel for, collapse(2), schedule(static)
2048	2	20	4.2358135	11.17	56	parallel for, collapse(2), schedule(static)
2048	3	1	45.636704	1	100	parallel for, schedule(static) + atomic (checksum)
2048	3	2	28.33491256	1.61	80	parallel for, schedule(static) + atomic (checksum)
2048	3	4	10.93567567	4.17	104	parallel for, schedule(static) + atomic (checksum)
2048	3	8	5.638275	8.09	101	parallel for, schedule(static) + atomic (checksum)
2048	3	12	5.698069111	8.01	67	parallel for, schedule(static) + atomic (checksum)
2048	3	16	4.366039889	10.46	65	parallel for, schedule(static) + atomic (checksum)
2048	3	20	4.410729889	10.35	51	parallel for, schedule(static) + atomic (checksum)
2048	4	1	44.96729211	1	100	parallel for, schedule(static) + critical (checksum)
2048	4	2	28.14964567	1.6	80	parallel for, schedule(static) + critical (checksum)
2048	4	4	10.91655878	4.12	103	parallel for, schedule(static) + critical (checksum)
2048	4	8	5.773897	7.79	97	parallel for, schedule(static) + critical (checksum)
2048	4	12	5.496331889	8.18	68	parallel for, schedule(static) + critical (checksum)
2048	4	16	4.341771444	10.35	65	parallel for, schedule(static) + critical (checksum)
2048	4	20	4.460456556	10.08	50	parallel for, schedule(static) + critical (checksum)
2048	5	1	46.68769289	1	100	task-based parallelism (blocked row decomposition, block_size=16)
2048	5	2	28.76937389	1.62	81	task-based parallelism (blocked row decomposition, block_size=16)
2048	5	4	10.90035156	4.28	107	task-based parallelism (blocked row decomposition, block_size=16)
2048	5	8	5.654234889	8.26	103	task-based parallelism (blocked row decomposition, block_size=16)
2048	5	12	4.943531889	9.45	79	task-based parallelism (blocked row decomposition, block_size=16)
2048	5	16	3.809861111	12.26	77	task-based parallelism (blocked row decomposition, block_size=16)
2048	5	20	4.116674222	11.34	57	task-based parallelism (blocked row decomposition, block_size=16)
2048	6	1	46.71265044	1	100	parallel for, schedule(static) + simd (inner k-loop vectorization)

2048	6	2	28.74152933	1.63	82	parallel for, schedule(static) + simd (inner k-loop vectorization)
2048	6	4	10.88477111	4.29	107	parallel for, schedule(static) + simd (inner k-loop vectorization)
2048	6	8	5.606167778	8.33	104	parallel for, schedule(static) + simd (inner k-loop vectorization)
2048	6	12	5.513533444	8.47	71	parallel for, schedule(static) + simd (inner k-loop vectorization)
2048	6	16	4.380005222	10.66	67	parallel for, schedule(static) + simd (inner k-loop vectorization)
2048	6	20	4.279193667	10.92	55	parallel for, schedule(static) + simd (inner k-loop vectorization)

Performance Analysis Discussion

1. Static vs Dynamic Scheduling

Static scheduling was the most appropriate strategy for this matrix multiplication workload because the computation is highly regular and evenly distributed. Each thread processes approximately the same number of rows and performs an identical number of arithmetic operations. Since there is no load imbalance, static scheduling minimizes runtime overhead and avoids the extra scheduling cost associated with dynamic scheduling. Therefore, static scheduling achieved stable and efficient performance across all matrix sizes.

2. Efficiency Drop with Increasing Threads

Efficiency decreases as thread count increases due to hardware and synchronization limitations. While speedup improves significantly up to 8 threads, efficiency drops beyond that point. This occurs because:

- Memory bandwidth becomes a bottleneck as more threads access shared data.
- Cache contention increases.
- Synchronization overhead grows with thread count.
- According to Amdahl's Law, serial portions of the program limit maximum achievable speedup.

At higher thread counts (12–16), diminishing returns are observed due to these architectural constraints rather than algorithmic inefficiency.

However, some configurations show efficiency slightly above 100%, indicating superlinear speedup. This is likely due to improved cache locality when the workload is divided among threads, reducing memory access latency compared to the serial case. Minor contributions may also come from measurement variability and CPU frequency scaling.

3. Atomic vs Critical for Checksum

The atomic implementation (Mode 3) consistently outperformed the critical implementation (Mode 4). Atomic operations provide hardware-level synchronization for simple memory updates, resulting in lower overhead. In contrast, critical sections rely on lock-based synchronization, which serializes execution and increases waiting time among threads. Therefore, atomic was faster while maintaining correctness for the checksum computation.

4. Tasks vs Loop-Based Parallelism

Task-based parallelism (Mode 5) showed stronger benefits for larger matrices (especially N = 2048). For smaller matrices (e.g., N = 512), task creation overhead reduced its advantage compared to loop-based parallelism. However, for large workloads, blocked row decomposition improved cache locality and load distribution, allowing tasks to achieve higher speedups at 16 threads. Thus, tasks are more beneficial when the computation granularity is sufficiently large to amortize scheduling overhead.

5. SIMD Optimization

SIMD optimization improved performance by vectorizing the inner k-loop. This enabled multiple multiply-add operations to be executed simultaneously within a single core using CPU vector units. SIMD improved single-thread performance and contributed to overall speedup when combined with multi-threading. However, SIMD operates within one core and does not replace thread-level parallelism. The best performance was achieved when SIMD and multi-threading were used together.

Output:

Matrix Size = 512, Thread 1

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.239655
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.247014
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.233342
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.244495
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.238830
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.234224
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.232882
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 512, Thread 2

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.244501
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.126943
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.124031
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.122107
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.128060
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.127422
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.122289
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 512, Thread 4

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.225355
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.067675
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.068943
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.065485
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.066713
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.070150
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.068891
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 512, Thread 8

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.249752
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.036968
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.036357
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.036190
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.047696
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.036192
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.052813
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 512, Thread 12

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.244915
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.027524
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.031594
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.028761
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.028953
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.028969
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.028494
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 512, Thread 16

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.243468
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.019716
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.023350
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.020595
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.027199
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.029714
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.022252
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 512, Thread 20

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 0
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.236323
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 1
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.017105
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 2
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.021615
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 3
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.021840
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 4
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.035397
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 5
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.034530
MaxC = 393467.072912
sumC = 67072852161.869759
checksum = 13115530876
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 512 6
c[0] =131583.221138
c[last] =393467.072912
Kernel Time = 0.034725
MaxC = 393467.072912
sumC = 67072852161.869774
checksum = 13115530876
```

Matrix Size = 1024, Thread 1

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.892687
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.904469
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.932116
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.931444
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.916949
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

```
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 5
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.900394
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 1024 6
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.905492
MaxC = 1573317.601707
sumC = 1072606412921.713501
checksum = 52429597478
```

Matrix Size = 1024, Thread 2

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 6
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.005586
MaxC = 1573317.601707
sumC = 1072606412921.713501
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 5
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.999091
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.995188
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.996656
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.000556
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

```
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.000556
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.004505
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.914925
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

Matrix Size = 1024, Thread 4

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.910829
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.531428
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.526570
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.530240
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.529521
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 5
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.534727
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 1024 6
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.530507
MaxC = 1573317.601707
sumC = 1072606412921.713501
checksum = 52429597478
```

Matrix Size = 1024, Thread 8

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.883780
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.284042
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.315858
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.282956
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.285166
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

```
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 5
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.290890
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 1024 6
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.282254
MaxC = 1573317.601707
sumC = 1072606412921.713501
checksum = 52429597478
```

Matrix Size = 1024, Thread 12

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.926129
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.304941
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.310078
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.296922
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.312317
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

Matrix Size = 1024, Thread 16

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.942134
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.242439
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.249792
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.237189
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.237034
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 5
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.190796
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 1024 6
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.240689
MaxC = 1573317.601707
sumC = 1072606412921.713501
```

Matrix Size = 1024, Thread 20

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 0
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 1.893762
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 1
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.228272
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 2
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.265887
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 3
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.239776
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 4
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.235861
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
```

```
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 5
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.200735
MaxC = 1573317.601707
sumC = 1072606412921.713379
checksum = 52429597478
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 1024 6
c[0] =525311.134616
c[last] =1573317.601707
Kernel Time = 0.229130
MaxC = 1573317.601707
sumC = 1072606412921.713501
checksum = 52429597478
```

Matrix Size = 2048, Thread 1

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 0
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 55.846782
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 53.404852
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 53.081930
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 53.948189
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 56.583151
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 51.949665
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 1 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 53.031887
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

Matrix Size = 2048, Thread 2

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 0
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 53.427541
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 27.854380
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 28.614192
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 26.219904
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 27.298842
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 28.714439
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 2 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 28.497522
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

Matrix Size = 2048, Thread 4

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 0
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 51.041815
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 11.095747
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 11.088618
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 10.788971
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 12.352058
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

```
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 10.446625
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 4 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 10.478046
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.667510
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.470123
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

Matrix Size = 2048, Thread 8

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.583109
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 6.006949
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 6.505279
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 6.080317
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 0

c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 66.312509
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.783252
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.886657
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 8 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.622589
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

Matrix Size = 2048, Thread 12

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 0
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 49.881965
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.647136
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.396794
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.092657
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.692575
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

```
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.447978
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 12 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.358655
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

Matrix Size = 2048, Thread 16

```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.518216
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 2048 0
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 54.152539
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.184270
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.425221
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

Matrix Size = 2048, Thread 20

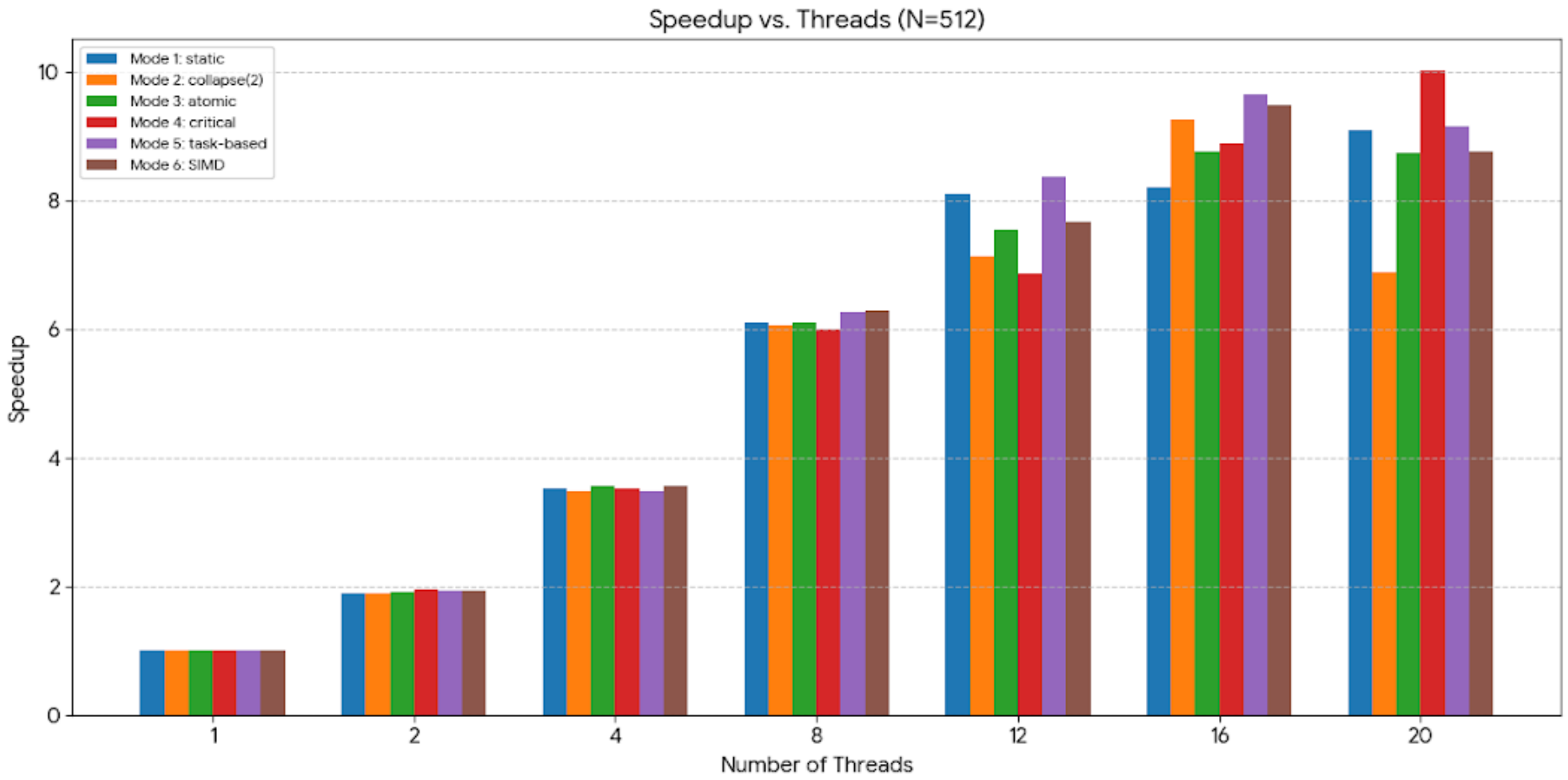
```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.051788
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 3.636035
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 16 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 5.121976
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

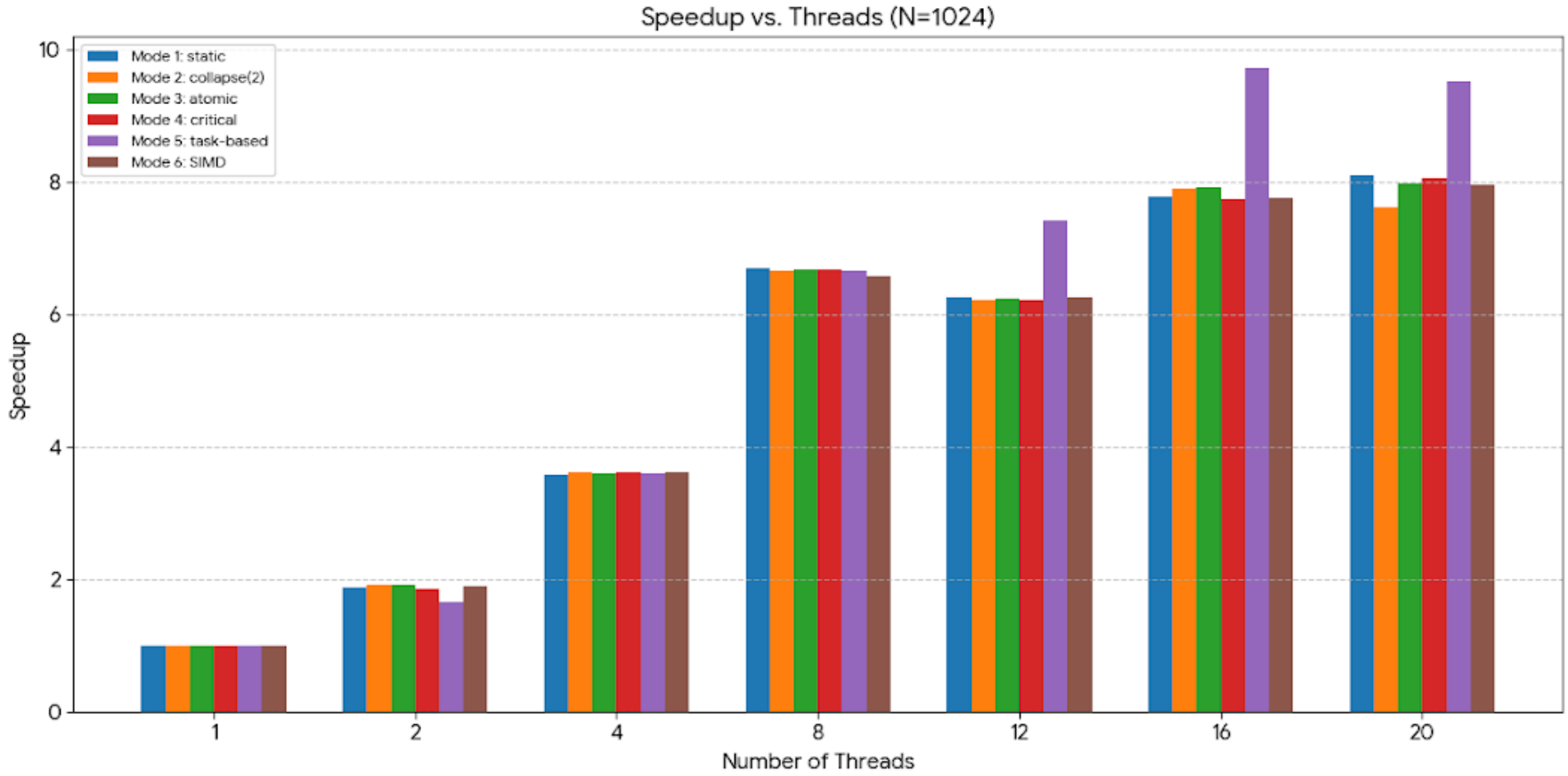
```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 5
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 3.857129
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 6
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.214771
MaxC = 6292466.572900
sumC = 17157232162900.759766
checksum = 209728603465
```

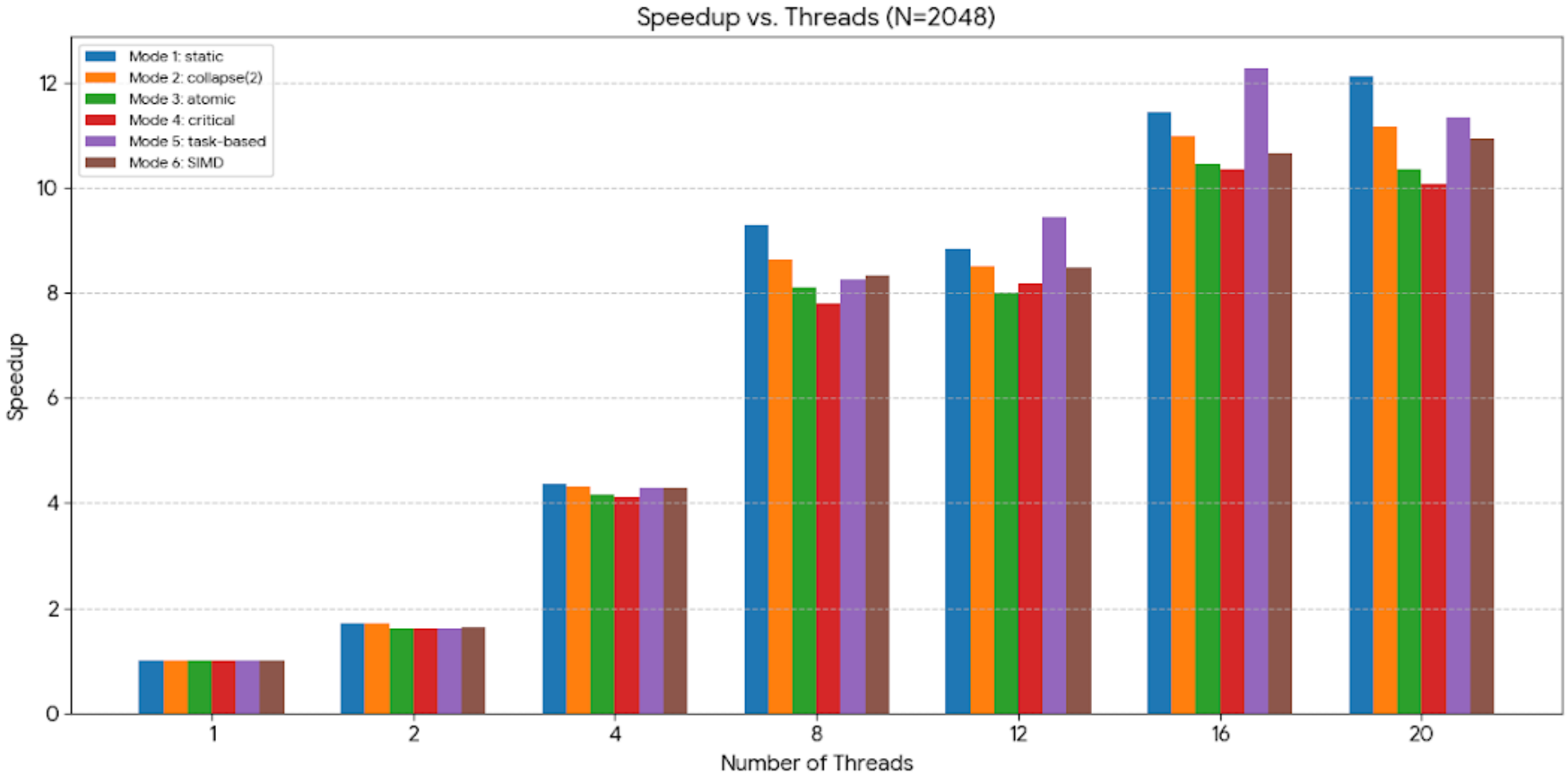
```
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 0
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 51.684071
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 1
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 3.805740
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 2
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 4.202358
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 3
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 3.855870
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
[y5mq2@remotelabm35 Parallel-Programming]$ setenv OMP_NUM_THREADS 20 ; ./a1 2048 4
c[0] =2099199.048034
c[last] =6292466.572900
Kernel Time = 3.809802
MaxC = 6292466.572900
sumC = 17157232162900.751953
checksum = 209728603464
```

Graphs:

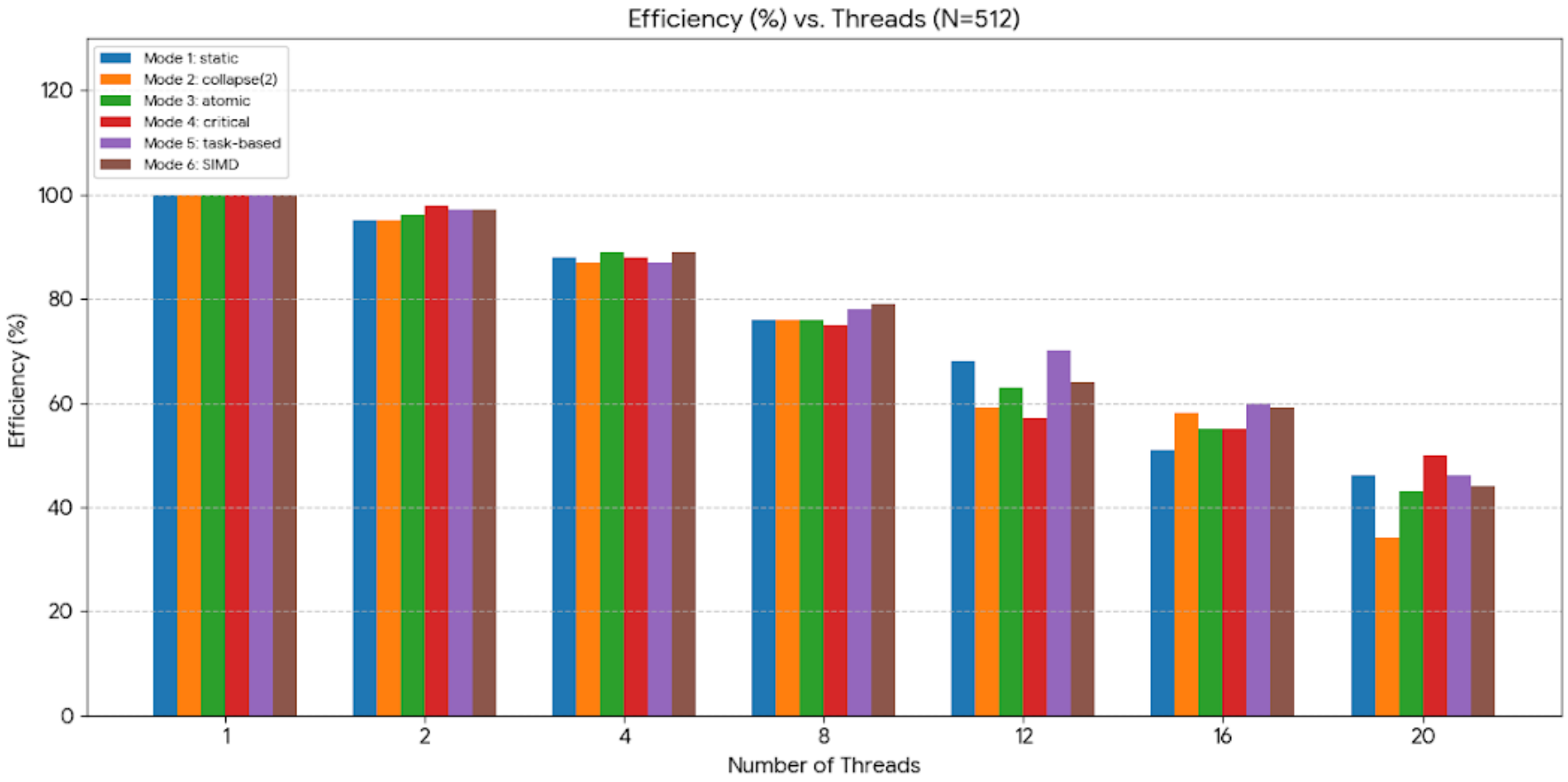
Speed up vs Threads

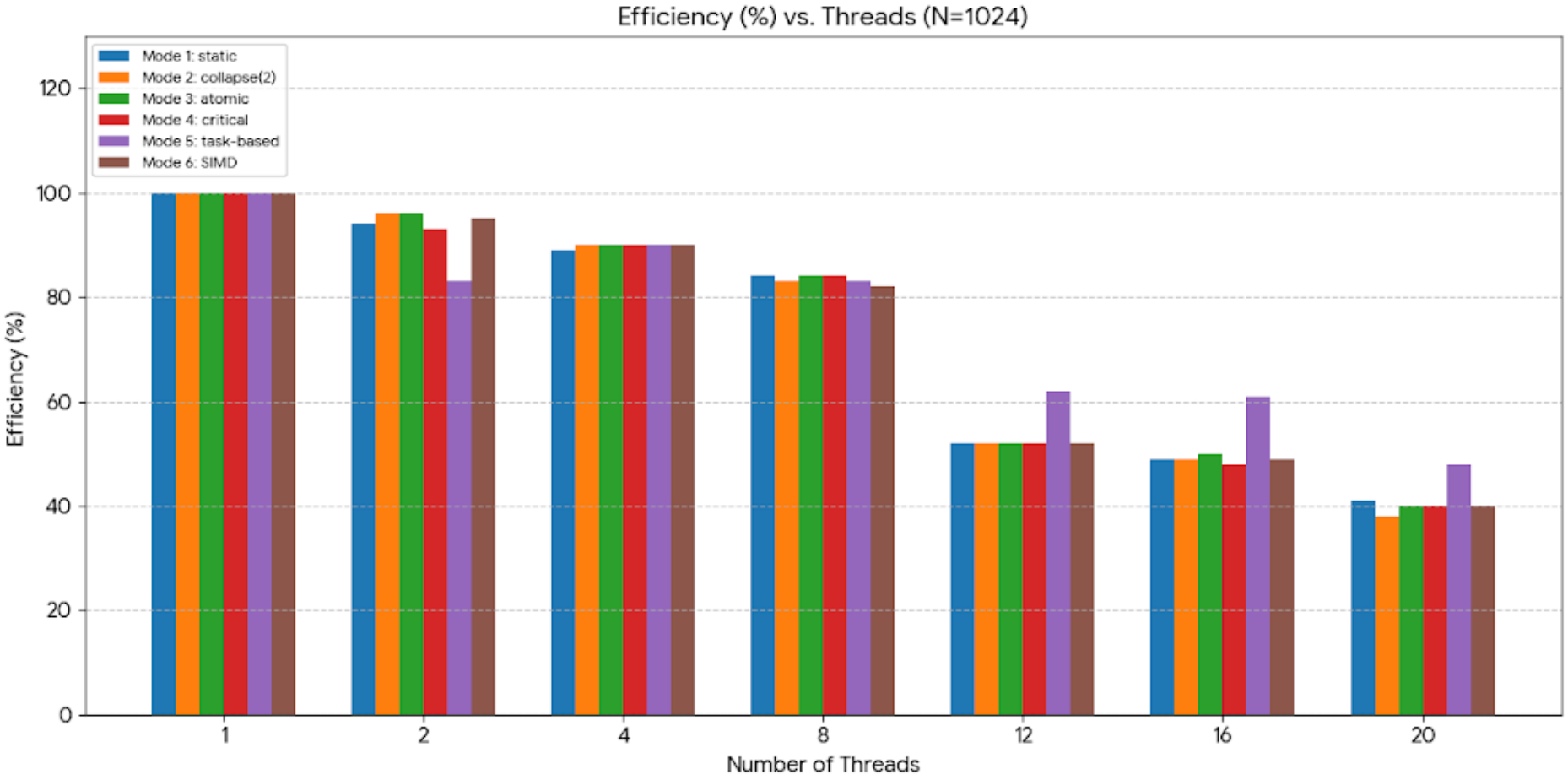


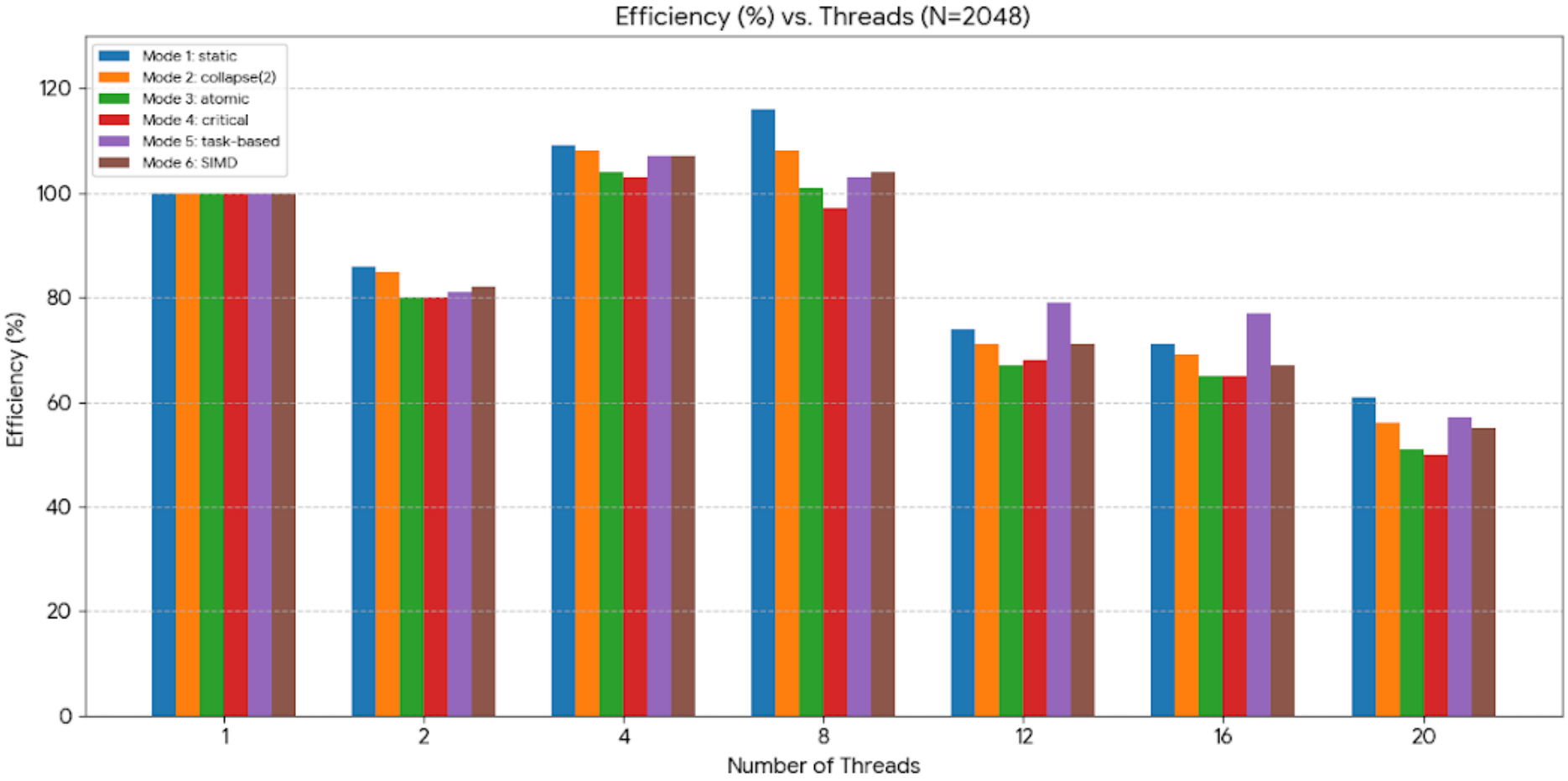

Speedup vs. Threads (N=1024)
Mode 1: static
Mode 2: collapse(2)
Mode 3: atomic
Mode 4: critical
Mode 5: task-based
Mode 6: SIMD
Speedup
0
2
4
6
8
10
1
2
4
8
12
16
20
Number of Threads


Speedup vs. Threads (N=2048)
Mode 1: static
Mode 2: collapse(2)
Mode 3: atomic
Mode 4: critical
Mode 5: task-based
Mode 6: SIMD
Speedup
0
2
4
6
8
10
12
1
2
4
8
12
16
20
Number of Threads

Efficiency vs Threads

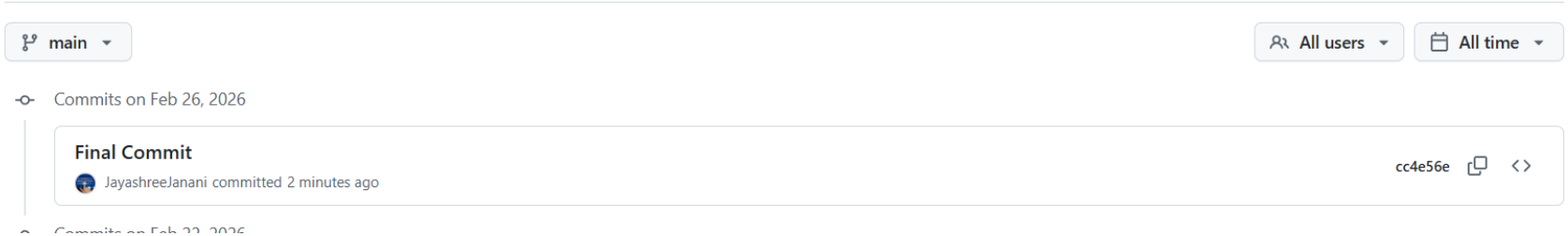



Efficiency (%) vs. Threads (N=1024)
Mode 1: static
Mode 2: collapse(2)
Mode 3: atomic
Mode 4: critical
Mode 5: task-based
Mode 6: SIMD
Efficiency (%)
0
20
40
60
80
100
120
1
2
4
8
12
16
20
Number of Threads


Efficiency (%) vs. Threads (N=2048)
Mode 1: static
Mode 2: collapse(2)
Mode 3: atomic
Mode 4: critical
Mode 5: task-based
Mode 6: SIMD
Efficiency (%)
0
20
40
60
80
100
120
1
2
4
8
12
16
20
Number of Threads

Git Repository:

https://github.com/JayashreeJanani/Parallel-Programming



Commits on Feb 22, 2026

Step 8: SIMD and threads-Complete JayashreeJanani committed 3 days ago	a27e710
Step 7 - Mode 4 Task Scheduling- complete JayashreeJanani committed 3 days ago	1857bba
Step 6- Mode 3 atomic and critical synchroniztion JayashreeJanani committed 3 days ago	99f487c
Step5:Mode2(Collapse(2))-complete JayashreeJanani committed 3 days ago	53f34bd

Commits on Feb 21, 2026

Completion of Step 4: Mode 1(threads +scheduling) JayashreeJanani committed 4 days ago	672f1c8
Completion of Step 3: Mode 0: Serial baseline JayashreeJanani committed 4 days ago	2ea2d10
Completion of Step2 JayashreeJanani committed 4 days ago	ea7e828
Step 2: Initialization of matrices JayashreeJanani committed 4 days ago	92cb937
Step 1B: Basic safety check for N JayashreeJanani committed 5 days ago	e2fdea5

Step 1A: Extracting N and mode
JayashreeJanani committed 5 days ago — a446ec7

Completion of Step 0 code
JayashreeJanani committed 5 days ago — b8fc4ba

Analyzing argc and argv
JayashreeJanani committed 5 days ago — fd62ad1

Baseline attempt before clean restart
JayashreeJanani committed 5 days ago — de0c106

Commits on Feb 19, 2026

19th February 2026 commit
JayashreeJanani committed last week — 0793100

Commits on Feb 12, 2026

Problem Statement prototype created
JayashreeJanani committed 2 weeks ago — e5299cc

Logic Building
JayashreeJanani committed 2 weeks ago — a7c04da